

PROFILE

MSc graduate in Applied Mathematics and Computation with a background in Engineering Physics and experience in mathematical modelling, numerical computation, machine learning and statistical analysis. Interested in data science, machine learning, quantitative analysis and analytical roles.

EDUCATION

- **University of Lisbon - Instituto Superior Técnico (IST)** Lisbon, Portugal
MSc in Applied Mathematics and Computation (Final grade: 17/20) 2025
Thesis topic: Analysis of Nonlinear PDEs (Grade: 19/20)
- **University of Lisbon - Instituto Superior Técnico (IST)** Lisbon, Portugal
BSc in Engineering Physics (Final grade: 15/20) 2019
Swiss-European Mobility exchange at École Polytechnique Fédérale de Lausanne (EPFL)

TECHNICAL SKILLS

- **Programming:** Python (proficient), C/C++ and R (academic experience).
- **Python:** NumPy, SciPy, pandas, scikit-learn, PyTorch, matplotlib.
- **Mathematics/Computing:** Mathematical and numerical modelling, dynamical systems, optimisation, machine learning, statistical/data analysis.
- **Tools:** Mathematica, LaTeX, Git.

RESEARCH EXPERIENCE

- **Research Assistant at Instituto Superior Técnico** Lisbon, Portugal
Ergodicity breaking in quantum many-body theory 2026 – Present
 - Developed Python-based numerical simulations of constrained dynamical systems using Monte Carlo and exact diagonalization methods; analyzed large computational datasets to identify signatures of phase transition.
- **Summer Research Assistant at Aalto University (ASci Program)** Helsinki, Finland
Equivariance properties of Graph Neural Networks (GNNs) 2023
 - Developed analytical results concerning equivariance properties of Graph Neural Networks.
- **Summer Research Assistant at Instituto Gulbenkian de Ciência (IGC)** Oeiras, Portugal
Evolution of cancer cell populations 2017
 - Developed and analysed an ODE mutation-selection model of cancer-cell population dynamics.

SELECTED PROJECTS

- **Mathematical Foundations of Machine Learning:** Wrote a set of notes covering the foundations of Learning Theory, Machine Learning and Kernel Methods with implementations in Python (repo).
- **Manifold Learning:** Applied manifold learning and nonlinear optimization techniques to dimensionality-reduction problems, with implementations in Python (repo).
- **Feature Selection:** Implemented information-theoretic feature-selection methods in Python and evaluated their effectiveness for dimensionality reduction and predictive modelling (repo).
- **Collaborative Filtering:** Developed a movie recommendation system using matrix-factorisation-based collaborative filtering, implementing model training and evaluation in Python (repo).

TEACHING EXPERIENCE

- **Invited Assistant Professor at ISCTE – Instituto Universitário de Lisboa** Lisbon, Portugal
Taught Calculus I, Calculus II and Linear Algebra to first-year Economics students, including classes taught in English. 2024, 2025
- **Private Tutor - Mathematics and Physics** Lisbon, Portugal
Individual instruction in mathematics and physics to high-school and undergraduate students. 2015 – 2023

ADDITIONAL TRAINING

- **Quantum Landscapes International School:** Paris, France, 2026.
- **LxMLS22 - Lisbon Machine Learning Summer School:** Lisbon, Portugal, 2022.

LANGUAGE SKILLS

- **Languages:** Portuguese (Native); English (Fluent); Spanish (Basic).